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```
FUNCTION inputReadings(sensors, readings, matrix)
    FOR i = 0 TO sensors - 1
        PRINT "Sensor " + (i+1) + ":"
        FOR j = 0 TO readings - 1
            INPUT matrix[i][j]
        NEXT j
    NEXT i
ENDFUNCTION

FUNCTION analyzeSensor(sensorNum, readingsCount, sensorRow)
    total = 0
    max = sensorRow[0]
    min = sensorRow[0]
    critical = 0
    warning = 0
    current_streak = 0
    max_streak = 0

    FOR j = 0 TO readingsCount - 1
        temp = sensorRow[j]
        total = total + temp

        IF temp > max THEN max = temp
        IF temp < min THEN min = temp

        IF temp >= 40 THEN critical = critical + 1
        IF temp >= 35 AND temp <= 39 THEN
            warning = warning + 1
```

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```
IF temp >= 35 THEN
    current_streak = current_streak + 1
    IF current streak > max_streak THEN
        max_streak = current_streak.
    ELSE
ELSE
    current_streak = 0
ENDIF
ENDIF
NEXT j

avg = total / readings Count.
PRINT "Sensor " + SensorNum + ": Total=" + total + "Avg="
    + avg + "Max= + max + "Min=" + min
PRINT "    Critical=" + critical + " Warning=" + warning +
    "Longest hot streak = " + max_streak
ENDFUNCTION.

FUNCTION findHighestAverage (sensors, readings, matrix)
    highest_avg = -1.0
    best_sensor = 1

    FOR i = 0 TO sensors - 1
        total = 0
        FOR j = 0 TO readings - 1
            total = total + matrix[i][j]
    NEXT j
```

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        avg = total // readings

        IF avg > highest_avg THEN
            highest_avg = avg
            best_sensor = i + 1
        ENDIF
    NEXT i
    RETURN best_sensor
ENDFUNCTION

START
BEGIN
    INPUT sensors, readings
    Let matrix[sensors][readings]

    CALL inputReadings(sensors, readings, matrix)

    PRINT "\n Sample output:"
    FOR i = 0 TO sensors - 1
        // Pass a single row (1D array) for individual
        // analysis.
        CALL analyzeSensor(i+1, readings, matrix[i])
    NEXT i.

best = CALL findHighestAverage(sensors, readings, matrix)
```

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```
// Recalculate avg for the winning sensor to print
best_total = 0
FOR j = 0 TO readings - 1
    best_total = best_total + matrix[best-1][j]
NEXT j
best_avg = best_total / readings.

PRINT "\n Highest average: Sensor " + best + "(" +
                best_avg + ")"
END.
```


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